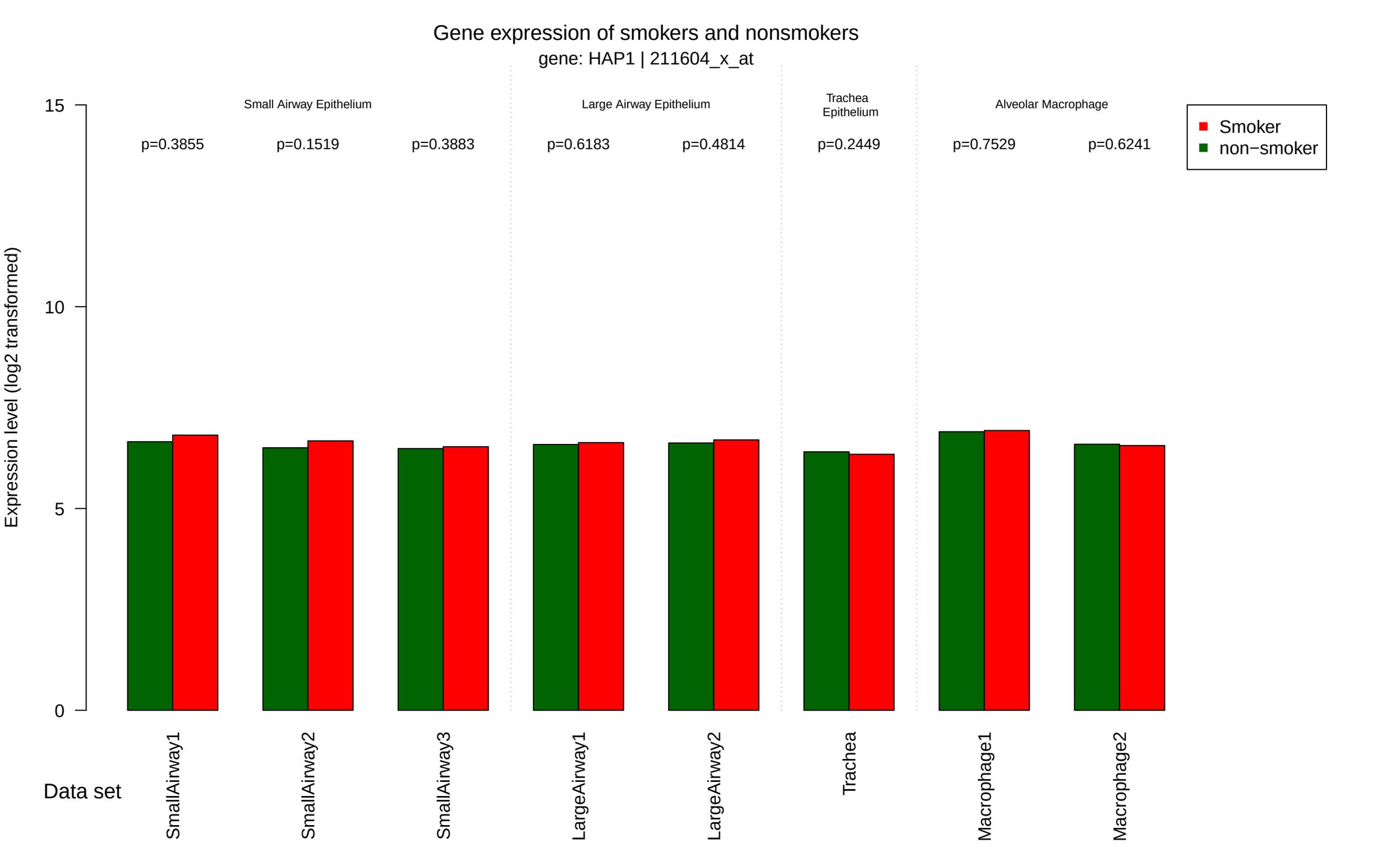
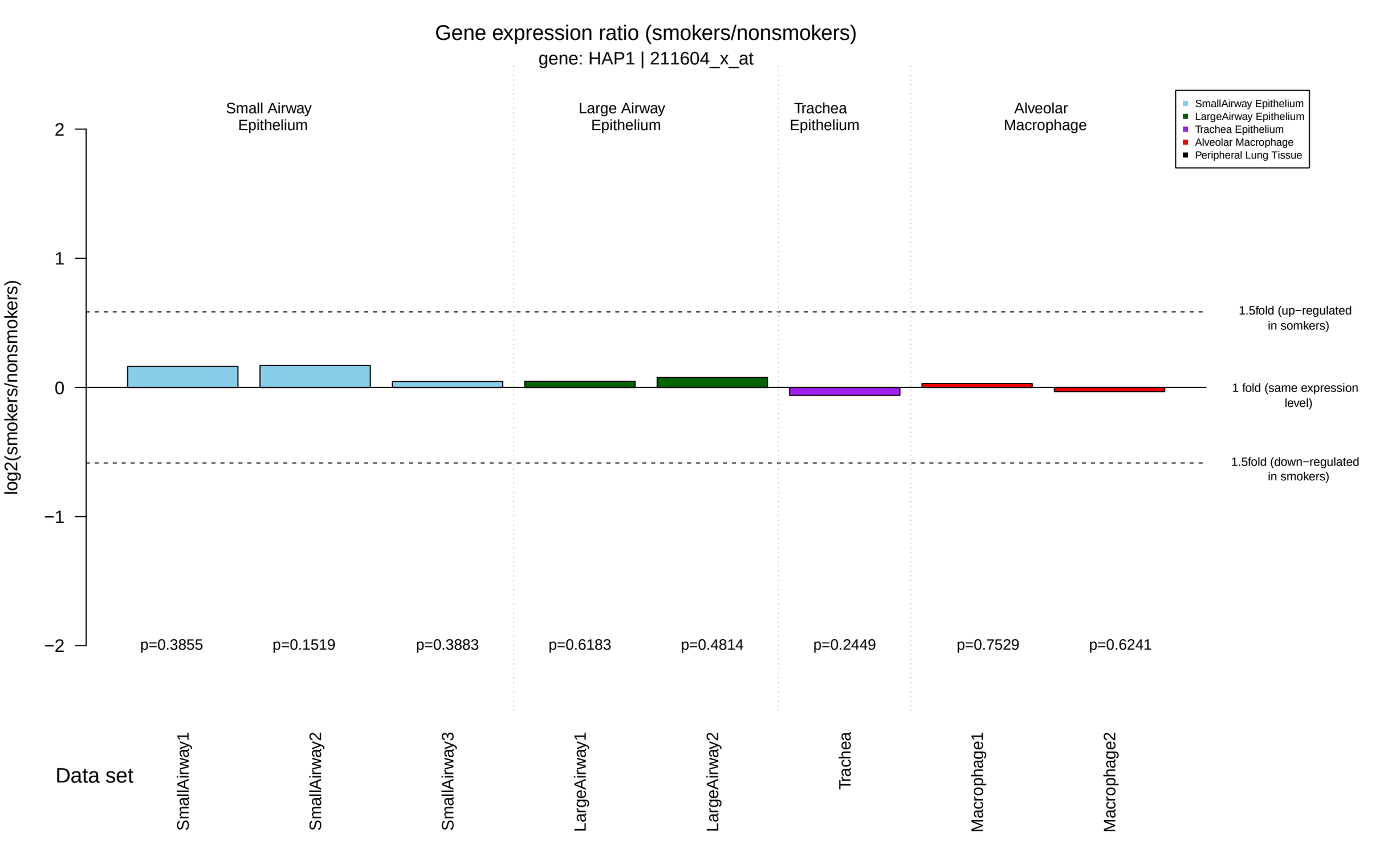
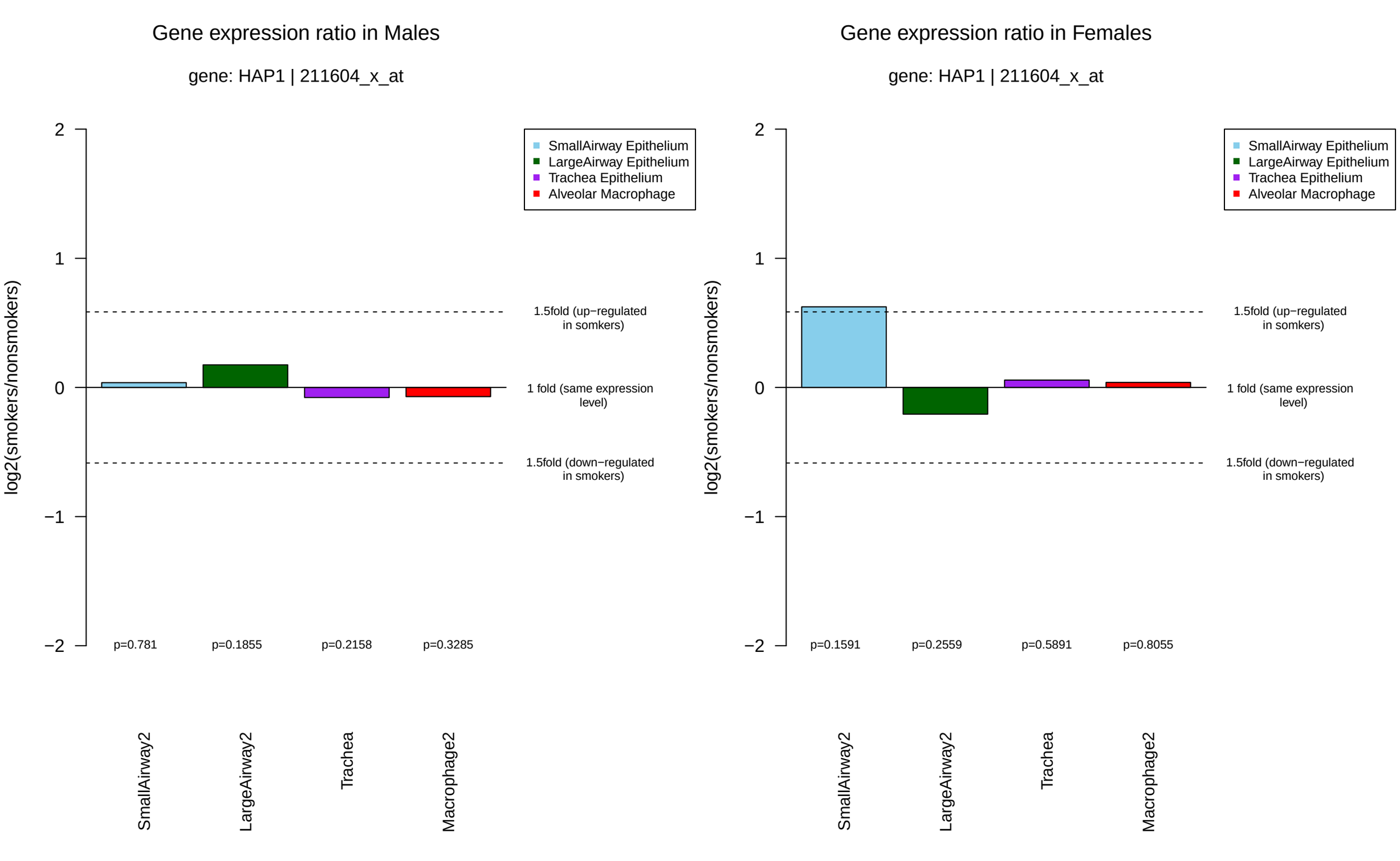
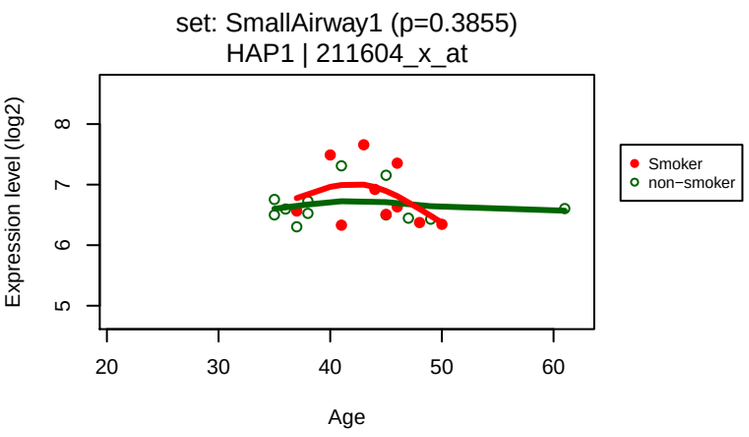
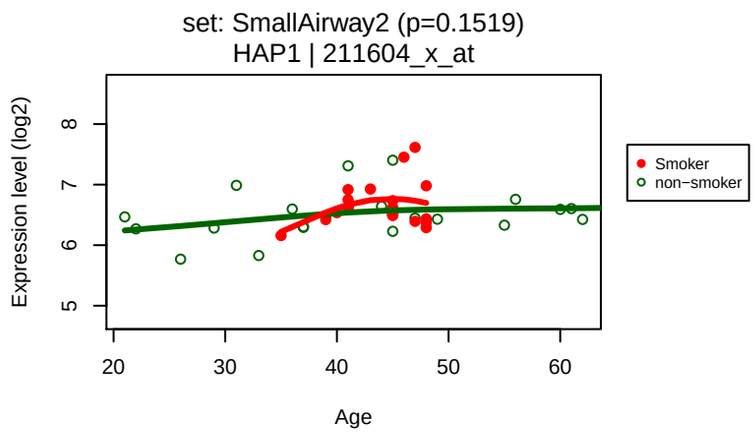
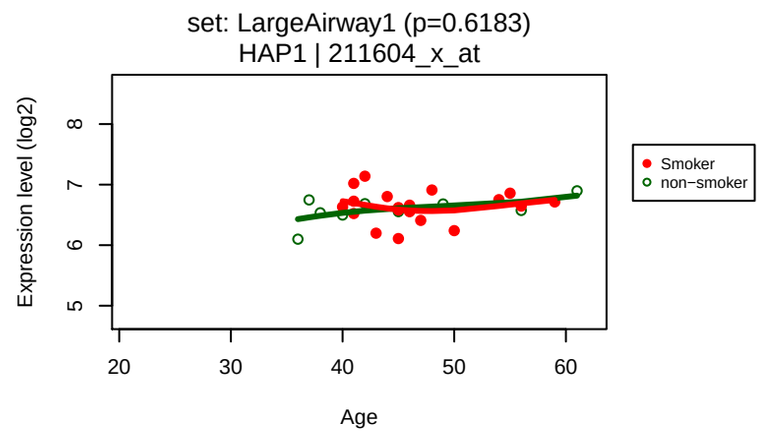

Gene expression of smokers and nonsmokers
gene: HAP1 | 211604_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3855
p=0.1519
p=0.3883
p=0.6183
p=0.4814
p=0.2449
p=0.7529
p=0.6241
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

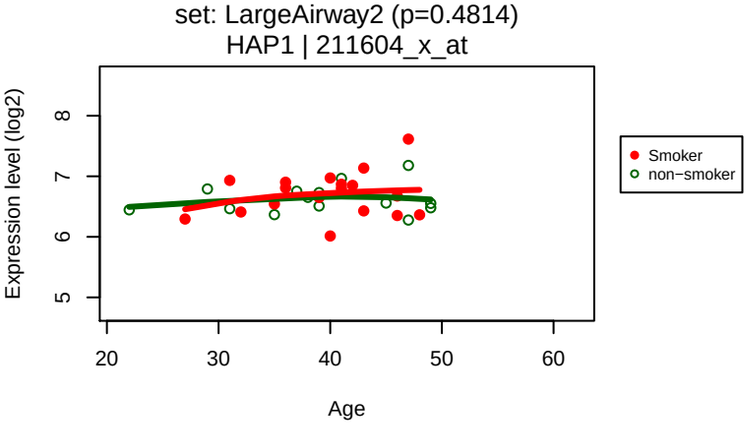

Gene expression ratio (smokers/nonsmokers)
gene: HAP1 | 211604_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3855
p=0.1519
p=0.3883
p=0.6183
p=0.4814
p=0.2449
p=0.7529
p=0.6241
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

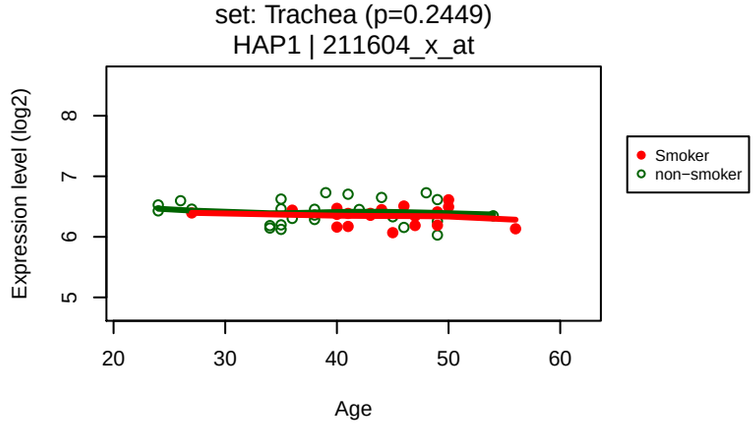

Gene expression ratio in Males
gene: HAP1 | 211604_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.781
p=0.1855
p=0.2158
p=0.3285
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HAP1 | 211604_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1591
p=0.2559
p=0.5891
p=0.8055
SmallAirway2
LargeAirway2
Trachea
Macrophage2

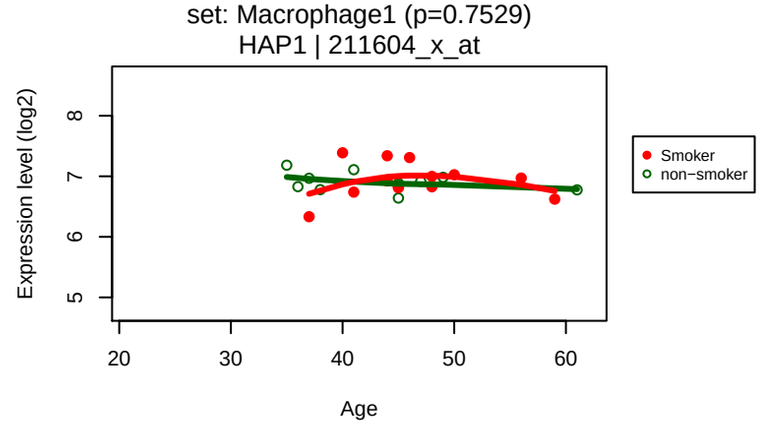

set: SmallAirway1 (p=0.3855)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker

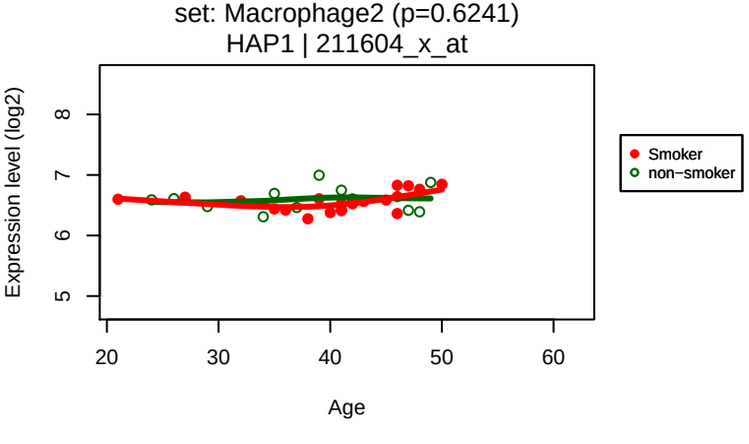

set: SmallAirway2 (p=0.1519)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker

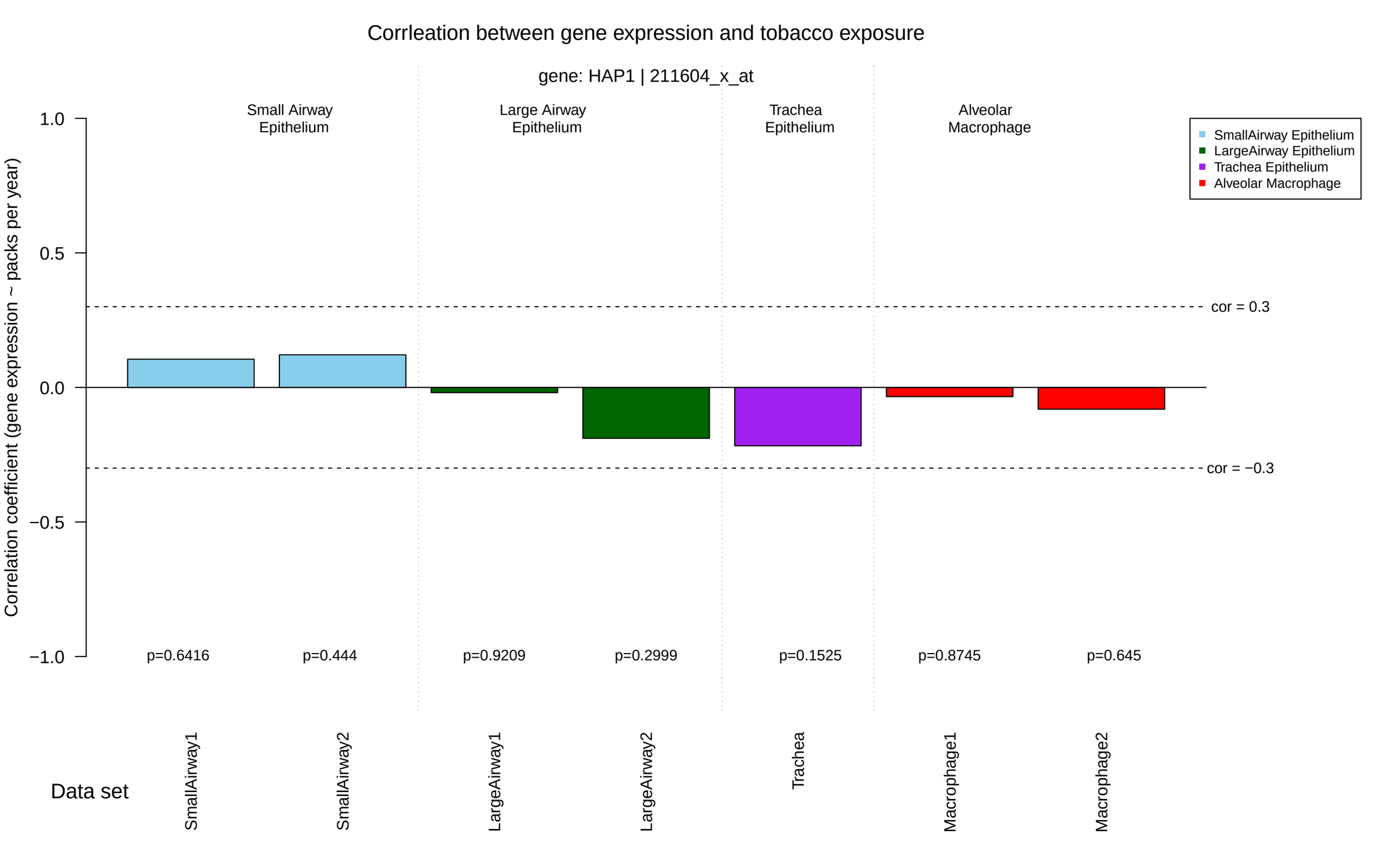

set: LargeAirway1 (p=0.6183)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4814)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2449)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7529)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6241)
HAP1 | 211604_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HAP1 | 211604_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6416
p=0.444
p=0.9209
p=0.2999
p=0.1525
p=0.8745
p=0.645
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2